

Redesigning a CT-Scan Department

A Simulation Study

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1 Introduction

Hospitals need to manage their CT scanners carefully, since scanners are expensive and patients have different levels of urgency. In the department, we study three types of patients use the same scanners: emergency patients, inpatients, and outpatients. During office hours two scanners are available. Outside office hours only one scanner is available, and it is only used for emergency patients and inpatients.

The hospital wants to redesign this department. Before doing that, they first want to know how the department performs right now. This report describes a simulation model of the current situation, and uses it to estimate the performance measures the hospital is interested in. We also do a simple analytical check to see whether two scanners should, in theory, be enough to handle the number of patients.

We use simulation because the system is too complex for a simple formula: there are three patient types, priority rules, a shared waiting room, and a weekly outpatient schedule. To deal with the fact that the simulation runs without ever really stopping, we use the batch means method to calculate 95% confidence intervals for our performance measures.

This report is organized as follows. Section 2 describes the current situation in words. Section 3 describes the simulation model: the events, the variables we track, and how we checked that the model works correctly. Section 4 explains how we calculate confidence intervals (Section 4.4) and presents the analytical capacity check (Section 4.1). Section 5 presents and discusses the results.

2 Problem Description

The department currently has two CT scanners. Both are open on weekdays from 8:00 to 16:00 (called office hours). Outside these hours, only one scanner stays open. A CT scan takes between 10 and 19 minutes (uniformly distributed), no matter the patient type. There are three types of patients:

- **Emergency patients** arrive unannounced, on average 24 per day. They always go first: if an emergency patient is waiting, they are scanned next, no matter who else is waiting.
- **Inpatients** are already in the hospital and can send a scan request at any time. On average 21 requests arrive during office hours and 6 outside office hours, per weekday. The arrival rate is not constant, but fluctuates according to a sine wave with peaks at 10:30 and 13:30. When a request comes in, the patient stays on the ward until the CT department calls for them. The department tries to have at most one inpatient in the waiting room at a time. After being called, it takes the patient 9 to 15 minutes (uniformly distributed) to arrive.
- **Outpatients** call ahead to book an appointment, which can be scheduled at the earliest the next day. On average 23 outpatients call on a weekday during office hours. Appointments are booked in 15-minute slots. At most 4 outpatients can be booked per hour in the morning and 3 per hour in the afternoon. If no slot is free in the same week, the patient goes on a waiting list, and is scheduled into next week on Friday evening. Each booked outpatient shows up with probability 0.84.

All patients wait in the same waiting room, which has 3 chairs. If a scanner is free when a patient arrives, they are scanned right away. If not, they wait for a chair if one is free. If all chairs are taken, they have to wait outside. Patients who do not show up are not counted. The waiting room is also used by pre-anesthesia screening (PAS) patients, but we leave them out of this simulation, as they are not part of this assignment.

Right now, the department uses a fixed weekly schedule: always two scanners during office hours and one outside office hours. Inpatients and outpatients are scanned in the order they arrive, except that emergency patients always go first. The goal of this report is to simulate this current setup and estimate the following performance measures, which the hospital wants insight into:

- CT scanner utilization, separately for office hours and outside office hours;
- access time for outpatients: the number of days between their request and their actual appointment;
- waiting time for emergency patients and outpatients: the time between arriving in the waiting room and the start of the scan;
- the fraction of patients who cannot find a free chair when they arrive;
- the fraction of inpatients who request a scan during office hours but are not scanned before 16:00 that same day.

Before building the simulation, Section 4.1 first checks, in a simple way, whether two scanners are even enough in theory. Sections 3 to 5 then describe the simulation model itself and the results we obtained from it.

3 Simulation Model

3.1 Entities and Assumptions

The model has three types of patients: emergency patients, inpatients, and outpatients. Each patient is represented by a simple object that stores its type, the time it requested a scan, and the time it arrived in the waiting room. Besides the patients, the model keeps track of the number of scanners currently busy, a queue of waiting emergency patients, a shared queue of waiting inpatients and outpatients (since these two types are scanned in order of arrival), a list of inpatients who have requested a scan but have not yet been called from the ward, a flag that says whether an inpatient is currently on their way, and the weekly outpatient schedule together with a waiting list of outpatients who could not be scheduled in the current week.

Inpatients are not put into a schedule like outpatients. Instead, an inpatient is called from the ward as soon as the CT waiting room contains no inpatients and none are already on their way; the model checks this condition every time a scan starts or ends. We also choose to schedule at most one outpatient per 15-minute slot, even though two scanners are available during office hours. This does not reduce the number of outpatients we can fit in, since it still respects the maximum number of outpatients allowed per hour (4 in the morning, 3 in the afternoon), and it spreads appointments out more evenly, which we expect to reduce unnecessary congestion in the waiting room.

We assume that a patient who does not show up for their outpatient appointment leaves no further trace in the system: they are not counted as having waited, and they do not occupy a chair. We also assume that a scanner is occupied for the full scan duration, with no setup or cleaning time in between scans, and that calling an inpatient from the ward does not depend on anything other than whether the waiting room already has an inpatient on the way or waiting.

3.2 Events

The model uses five types of events, plus one small routine ("call next inpatient") that is shared between two of them.

1. Request event. This represents a patient of any type requesting a scan, and always schedules the next request of the same type right away.

Algorithm 1 Request event

```

1: Create a new patient of the given type
2: if patient is Emergency then
3:   Send patient to the waiting room immediately (Arrival event)
4: else if patient is Inpatient then
5:   Add patient to the list of inpatients waiting on the ward
6: else if patient is Outpatient then
7:   Try to schedule patient in the earliest available slot
8: end if
9: Sample the time of the next request of this type
10: Schedule a new Request event at that time

```

2. Arrival event. This event represents a patient physically reaching the waiting room: immediately for emergency patients, after the call-and-travel delay for inpatients, and at their scheduled time for outpatients.

Algorithm 2 Arrival event

```

1: if patient is Outpatient and does not show up then
2:   return
3: end if
4: if a scanner is free then
5:   Start the scan immediately
6: else
7:   Add patient to the appropriate queue (emergency, or shared inpatient/outpatient queue)
8: end if
9: Record arrival time and whether a chair was available

```

3. Start-of-scan event. This event represents a scanner starting a scan.

Algorithm 3 Start-of-scan event

```

1: Sample scan duration  $\sim$  Uniform(10, 19)
2: Schedule End-of-scan event at current time + scan duration
3: if an inpatient is waiting on the ward and none is in the waiting room or already on their way then
4:   CALLNEXTINPATIENT
5: end if
6: Record the patient's waiting time

```

4. End-of-scan event. This event represents a scanner finishing a scan and, if possible, immediately starting the next one.

Algorithm 4 End-of-scan event

```

1: Free the scanner
2: if a scanner is available and a patient is waiting then
3:   Select the next patient (emergency first, otherwise the shared queue)
4:   Start that patient's scan
5: end if
6: if an inpatient is waiting on the ward and none is in the waiting room or already on their way then
7:   CALLNEXTINPATIENT
8: end if

```

Calling an inpatient is not a separate event, but a short routine used by the two events above:

Algorithm 5 CallNextInpatient()

```

1: Sample travel time  $\sim$  Uniform(9, 15)
2: Remove the next inpatient from the ward waiting list
3: Schedule an Arrival event for that patient at current time + travel time
4: Mark that an inpatient is currently on their way

```

5. Schedule-next-week event. This event happens once a week, on Friday evening, and represents the department planning next week's outpatient appointments.

Algorithm 6 Schedule-next-week event

- 1: Let L be the outpatients currently on the waiting list
 - 2: Schedule patients in L into next week's slots (at most one per slot, 4 per hour in the morning, 3 per hour in the afternoon)
 - 3: Schedule the next Schedule-next-week event, one week later
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3.3 Data collection

At the specific times mentioned earlier, the model records several key metrics. First, it tracks scanner utilization by logging the number of busy scanners over time, separated into office hours and outside office hours. Second, it records access time as the duration between an outpatient's initial request and their scheduled appointment. Third, it measures waiting time by tracking the period between a patient's arrival in the waiting room and the start of their scan for both emergency and outpatients. Fourth, it monitors waiting room capacity by noting whether or not an arriving patient found an empty chair. Finally, it checks inpatient scan success by recording whether an inpatient who requested a scan during office hours was successfully scanned before 16:00 on the same day.

4 Statistical Analysis

4.1 Analytical Capacity Check

Before building the simulation, we first check, with a simple calculation, whether two scanners should in theory be enough to handle the number of patients. The idea is straightforward: we estimate how much scanning time is needed on average, and compare that to how much scanning time is actually available.

A scan takes, on average, $\frac{10+19}{2} = 14.5$ minutes, since scan times are uniformly distributed between 10 and 19 minutes. We use this average scan time throughout this check.

We look at this separately for office hours and for outside office hours, and we do this over a full week rather than a single day, since the only scanner available outside office hours also has to deal with the entire weekend, not just weekday evenings. For emergency patients and inpatients, we assume the same average demand applies on every day of the week, including weekends, since the problem description only specifies rates for weekdays and this is also how we built the simulation model.

During office hours (8:00–16:00 on weekdays), two scanners are available, for 5 days a week, which gives $2 \times 8 \times 5 \times 60 = 4800$ minutes of scanning capacity per week. On average, $24 \times \frac{8}{24} = 8$ emergency patients, 21 inpatients, and $23 \times 0.84 = 19.32$ outpatients (after accounting for no-shows) arrive during office hours per weekday, giving $5 \times (8 + 21 + 19.32) = 241.6$ patients per week. At 14.5 minutes per scan, this requires about $241.6 \times 14.5 \approx 3503$ minutes of scanning per week.

Outside office hours, only one scanner is available, for the remaining $168 - 40 = 128$ hours of the week, giving $128 \times 60 = 7680$ minutes of capacity. Outside office hours, we expect $24 \times 7 - 40 = 128$ emergency patients and, since the same daily pattern of 27 inpatients (21+6) applies to all 7 days, $27 \times 7 - 21 \times 5 = 84$ inpatients per week. This gives $128 + 84 = 212$ patients, requiring about $212 \times 14.5 \approx 3074$ minutes of scanning per week.

Table 1: Expected weekly scanning demand versus available capacity.

	Demand (min/week)	Capacity (min/week)	Implied utilization
Office hours	3503	4800	73%
Outside office hours	3074	7680	40%

In both cases, the expected demand stays well below the available capacity, so two scanners during office hours and one scanner outside office hours should in theory be enough.

We also check whether there are enough appointment slots for outpatients. On a weekday, outpatients can be booked into at most 4 slots per hour in the morning and 3 per hour in the afternoon, giving $4 \times 4 + 3 \times 4 = 28$ available slots per day. Since on average 23 outpatients call per weekday, this leaves a margin of about 5 slots a day, which should be enough to also absorb outpatients who could not be scheduled in the same week.

Overall, this simple check suggests that the current setup – two scanners during office hours, one scanner outside office hours, and the current appointment slot limits – should be sufficient to handle the expected number of patients.

4.2 Verification

Before trusting the results from the full model, we check the simulation itself against a few cases where an analytical answer is known. To do this, we temporarily switched off inpatient and outpatient requests, leaving only emergency patients, and fixed the number of scanners to one at

all times. Under these settings, the model reduces to a single-server queue with Poisson arrivals, for which exact formulas exist.

M/G/1 check. With only emergency patients present, arrivals are Poisson with rate λ , scan times are Uniform(10, 19), and there is a single server. This is exactly an M/G/1 queue, so we can use the following formula to get the mean waiting time:

$$W_q = \frac{\lambda \mathbb{E}[S^2]}{2(1 - \rho)}, \quad \rho = \lambda \mathbb{E}[S], \quad (1)$$

where S is the scan time. With $\lambda = 24$ patients per day and $S \sim \text{Uniform}(10, 19)$, this gives $\rho = 0.242$ and a theoretical waiting time of $W_q = 2.385$ minutes. The simulated 95% confidence interval for the emergency waiting time under this same single-server, emergency-only setup is (2.339, 2.421) minutes, which contains the theoretical value. This gives us confidence that the core queueing logic of the simulation, the part that decides who gets the scanner next and how long they wait, is implemented correctly.

M/M/1 check. To check the engine against a second known case, we also replaced the Uniform(10, 19) scan time with an exponential distribution with the same mean (14.5 minutes), keeping a single server and emergency patients only. This gives an M/M/1 queue, for which

$$W_q = \frac{\rho}{\mu(1 - \rho)}, \quad \rho = \frac{\lambda}{\mu}, \quad (2)$$

with $\mu = 1/14.5$ scans per minute. This gives a theoretical $W_q = 4.621$ minutes. The simulated 95% confidence interval was (4.536, 4.747) minutes, again matching the theoretical value.

4.3 Data Collected

While the simulation runs, we keep track of the data needed for the five performance measures listed in Section 2: how busy the scanners are, how long outpatients wait for their appointment, how long patients wait in the waiting room, how often a patient cannot find a chair, and how often an inpatient is not scanned before 16:00. These are exactly the things we already described as being recorded in Section 3. We collect this data continuously while the simulation runs, and turn it into confidence intervals using the method described below.

4.4 Long Run Statistic

The simulation never really stops: it represents an ongoing process, week after week. Because of this, we cannot just run it once and treat the result as one clean observation. Instead, we use the *batch means method*. The idea is simple: we run one long simulation, but split it into many smaller pieces, called batches. Each batch gives us one value for each performance measure (for example, one batch might give a scanner utilization of 0.43). If we have enough batches, we can treat these batch values as roughly independent observations, and use them to build a normal 95% confidence interval, in the same way we would for any other sample of data.

To make sure we use enough batches, we need at least r batches, where r satisfies

$$r \geq \frac{t_{r-1, 1-\alpha/2}^2 S^2(r)}{\left(\frac{\delta}{1+\delta} \bar{X}(r)\right)^2}. \quad (3)$$

Here, $\bar{X}(r)$ and $S^2(r)$ are the sample mean and sample variance over the batches, $t_{r-1, 1-\alpha/2}$ comes from the t -distribution, α is the significance level, and δ is the precision we want: how close

the confidence interval should be to the mean, relative to the mean itself. We use $\alpha = 0.05$ and $\delta = 0.10$, meaning we want a 95% confidence interval that is within 10% of the estimated mean.

We chose $r = 150$ batches in advance, and then checked, for every performance measure, how many batches would actually be needed to reach this precision. These minimums are shown in Table 2. All of them are below 150, so 150 batches is enough for every performance measure we are interested in.

Table 2: Minimum number of batches needed for 10% precision.

Performance measure	Minimum batches needed
Scanner utilization office hours	1.13
Scanner utilization outside office hours	0.87
Access time for outpatients (days)	51.98
Waiting time for outpatients (minutes)	5.995
Waiting time for emergency patients (minutes)	2.463
Fraction of patients waiting without a chair	149.9
Fraction of inpatients not scanned before 16:00	56.54

The fraction of patients waiting without a chair needs by far the most batches, almost reaching our chosen $r = 150$. This makes sense: this event is rare, so the batch values for this measure vary a lot from one batch to the next, and we need more batches to average that variation out.

Before we can start collecting batches, the simulation needs some time to settle into a normal pattern, since it starts from an empty system. We looked at how the performance measures behaved over time and found that they became stable after about 4 weeks. We use this as our warm-up period and do not use any data from before this point. After the warm-up period, we split the rest of the simulation into batches of 6 weeks each. We use whole weeks for both the warm-up period and the batch length, since the start of a new week is a natural restart point for the system: that is exactly when new outpatient appointments are scheduled.

5 Results

Table 3 shows the 95% confidence interval for each performance measure, using $r = 150$ batches, a warm-up period of 4 weeks, batches of 6 weeks, and random seed 70.

Table 3: 95% confidence intervals for the performance measures.

Performance measure	95% confidence interval
Scanner utilization, office hours	(0.693, 0.703)
Scanner utilization, outside office hours	(0.417, 0.423)
Waiting time, outpatients (minutes)	(2.45, 2.54)
Waiting time, emergency patients (minutes)	(3.66, 3.75)
Access time, outpatients (days)	(1.84, 2.05)
Fraction of patients waiting without a chair	(0.0021, 0.0025)
Fraction of inpatients not scanned before 16:00	(0.038, 0.042)

Scanner utilization. The scanners are busy about 70% of the time during office hours, and about 42% of the time outside office hours. This matches well with the simple calculation in Section 4.1, where we estimated 73% for office hours and 40% for outside office hours. The fact that these two numbers are close gives us confidence that the simulation model behaves as expected, since the analytical check and the simulation use the same patient demand but a very different method to estimate utilization.

Waiting times. Outpatients wait, on average, between 2.45 and 2.54 minutes before their scan starts, and emergency patients wait between 3.66 and 3.75 minutes. These waiting times are both short, which makes sense: emergency patients are given priority, and outpatients have a pre-booked appointment, so they should rarely have to wait long unless the schedule is running behind. It may seem surprising that emergency patients wait slightly longer than outpatients on average, but this happens because emergency patients can arrive at any moment, including exactly when both scanners are busy, while outpatients are scheduled into slots that are deliberately spread out to avoid exactly that situation.

Access time for outpatients. On average, outpatients wait between 1.84 and 2.05 days between their call and their actual appointment. Since appointments can only be booked from the next day onwards, and there are slightly more appointment slots available than calls per day (28 slots against 23 calls on average, see Section 4.1), this short access time also matches what we would expect.

Waiting without a chair. Only about 0.21% to 0.25% of patients have to wait without finding a chair. This means the three chairs currently in the waiting room are almost always enough, and increasing this number is unlikely to meaningfully improve patient comfort under the current patient volumes.

Inpatients not scanned before 16:00. Between about 3.8% and 4.2% of inpatients who request a scan during office hours are not scanned before 16:00 that day. This is a small but not negligible fraction: roughly 1 in 25 such inpatients ends up being scanned in the evening instead of during the day, which the hospital described as undesirable.

Overall, the results line up well with the simple capacity check from Section 4.1: the system has enough capacity on average, but is busy enough during office hours (around 70%) that a small fraction of inpatients still get pushed past the 16:00 deadline. This is expected in any system with random arrivals, even when there is, on average, enough capacity: utilization does not need to reach 100% for occasional delays to occur.

6 Conclusion

In this report, we built a simulation model of the CT scan department as it currently operates, with two scanners during office hours and one scanner outside office hours, and used it to estimate how well the department is performing.

A simple calculation showed that, on average, the department has enough scanning capacity and enough appointment slots to handle the number of patients it currently receives. The simulation results support this: the scanners are busy a reasonable amount of the time (around 70% during office hours, 42% outside office hours), waiting times for emergency patients and outpatients are both short (a few minutes), outpatients get an appointment within about two days of calling, and almost every patient finds a free chair in the waiting room.

The one point worth highlighting to the hospital is that a small fraction of inpatients, about 4 in 100, are not scanned before the end of office hours on the day they request a scan, even though enough scanning capacity exists on average. This happens because patient arrivals are random: even a system that has enough capacity overall will occasionally have several patients need a scan around the same time, causing short, unavoidable delays near the end of the day.

In short, the current setup with two CT scanners appears to be sufficient for the patient volumes the department currently sees, and does not show signs of serious overcrowding or excessive waiting. Any future redesign should keep an eye on this small group of delayed inpatients, since they are the clearest sign of the system being stretched near the end of office hours.

A Appendix

Our simulation code can be found in the following repository: <https://github.com/MilanVeul/Mod8-StochasticSimulation>

We have put great efforts in organizing our code for readability and modularize it for future extensions or reuse.